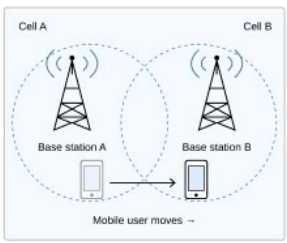
Handoff in Mobile Network

Handoff in Mobile Network

A moving mobile phone changes cells. Handoff lets its ongoing connection move to another base station.



In this experiment

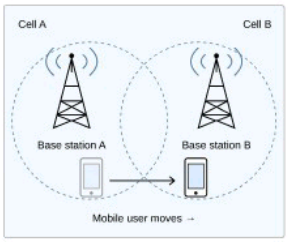
Follow the handoff process and learn which metrics help the network decide.

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What is Handoff?

What is Handoff?

Handoff is the process of transferring an ongoing call or connection from one base station to another when a mobile user moves between cells.



What changes?

The serving base station changes from A to B. The call or connection stays active.

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Why Do We Need Handoff?

Why Do We Need Handoff?

Handoff maintains continuous communication when a mobile phone moves out of one cell's coverage area into another.

Leaving Cell A → Entering Cell B

- **Signal strength**
  The signal from the first base station weakens as the phone moves away.
- **Connection quality**
  A better neighbouring base station helps maintain a reliable connection.
- **Uninterrupted calls**
  Transferring the connection helps avoid dropped calls as the user moves.

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Handoff Process and Metrics

Handoff Process and Metrics

1. Mobile phone moves away from the first base station.
2. Signal strength decreases.
3. The network detects a better neighbouring base station.
4. Connection transfers to the new base station.
5. Communication continues without interruption.

Metrics

- Signal Strength
- Signal Quality
- Distance to Neighbouring Cells
- Speed of Movement

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